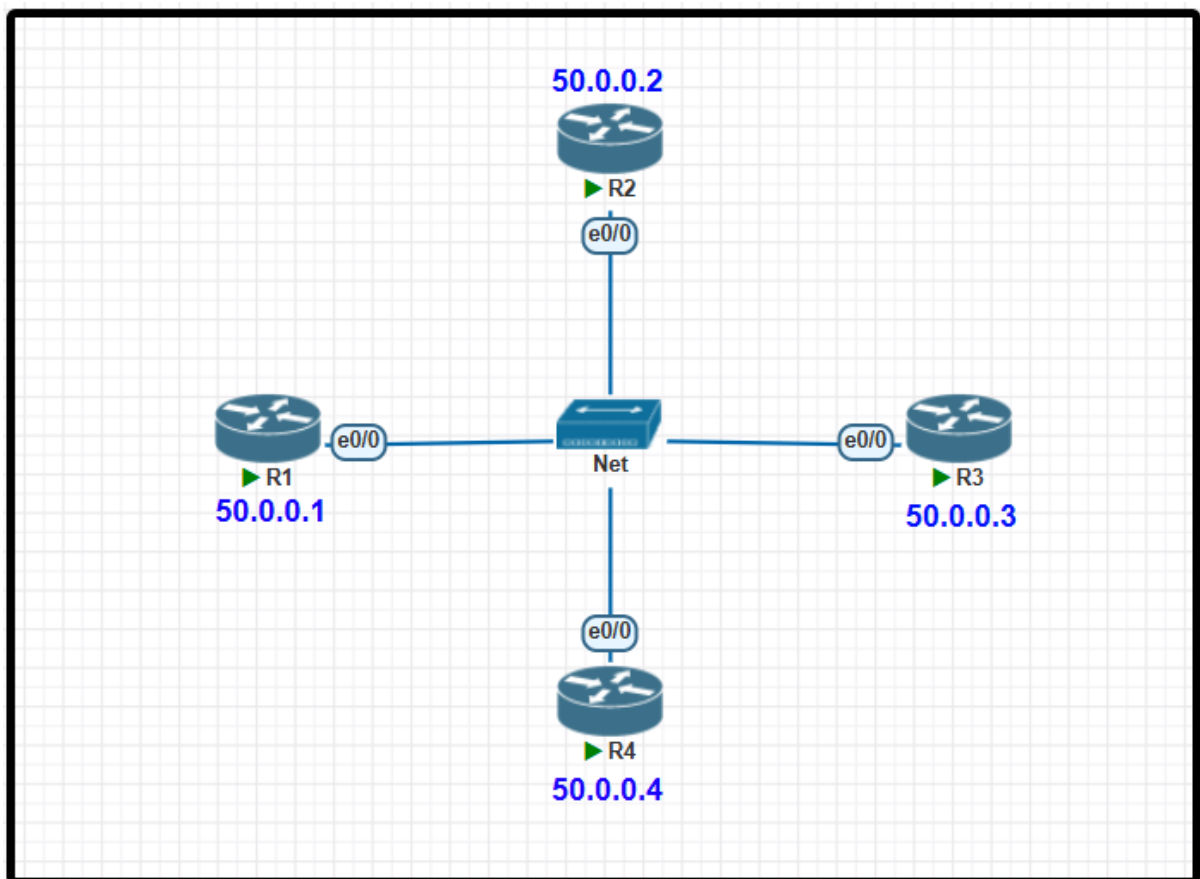


OSPF DR BDR BROTHERS



Step 1 : Assign ip address in all cisco routers

R1						
R1#sh ip int br						
Interface	IP-Address	OK?	Method	Status	Prot	
Ethernet0/0	50.0.0.1	YES	TFTP	up	up	
Ethernet0/1	unassigned	YES	TFTP	down	down	
Ethernet0/2	unassigned	YES	TFTP	down	down	
Ethernet0/3	unassigned	YES	TFTP	down	down	

R2						
R2#sh ip int br						
Interface	IP-Address	OK?	Method	Status	Prot	
Ethernet0/0	50.0.0.2	YES	TFTP	up	up	
Ethernet0/1	unassigned	YES	TFTP	down	down	
Ethernet0/2	unassigned	YES	TFTP	down	down	
Ethernet0/3	unassigned	YES	TFTP	down	down	

R3						
R3#sh ip int br						
Interface	IP-Address	OK?	Method	Status	Prot	
Ethernet0/0	50.0.0.3	YES	TFTP	up	up	
Ethernet0/1	unassigned	YES	TFTP	down	down	
Ethernet0/2	unassigned	YES	TFTP	down	down	
Ethernet0/3	unassigned	YES	TFTP	down	down	

R4						
R4#sh ip int br						
Interface	IP-Address	OK?	Method	Status	Prot	
Ethernet0/0	50.0.0.4	YES	TFTP	up	up	
Ethernet0/1	unassigned	YES	TFTP	down	down	
Ethernet0/2	unassigned	YES	TFTP	down	down	
Ethernet0/3	unassigned	YES	TFTP	down	down	

Step 2 : Enable ospf protocol in all cisco router

```
R4
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#
R4(config)#router ospf 4
R4(config-router)#
R4(config-router)#router-id 4.4.4.4
R4(config-router)#
R4(config-router)#network 50.0.0.4 0.255.255.255 area 1
R4(config-router)#
R4(config-router)#exit
R4(config)#

R3
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#
R3(config)#router ospf 3
R3(config-router)#
R3(config-router)#router-id 3.3.3.3
R3(config-router)#
R3(config-router)#network 50.0.0.3 0.255.255.255 area 1
R3(config-router)#
R3(config-router)#exit

R2
R2(config)#router ospf 2
R2(config-router)#
R2(config-router)#router-id 2.2.2
*Sep 24 03:26:06.669: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R2(config-router)#router-id 2.2.2.2
R2(config-router)#
R2(config-router)#network 50.0.0.2 0.255.255.255 area 1
R2(config-router)#exit

R1
*Sep 24 03:26:42.600: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R1(config)#router ospf 1
R1(config-router)#
R1(config-router)#router-id 1.1.1.1
R1(config-router)#
R1(config-router)#network 50.0.0.1 0.255.255.255 area 1
R1(config-router)#exit
R1(config)#exit
R1#
```

Step 3 : check the DR and BDR ip in the Neighbor table

```
R1
R1#sh ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
2.2.2.2          1     2WAY/DROTHER    00:00:33    50.0.0.2       Ethernet0/0
3.3.3.3          1     FULL/BDR        00:00:38    50.0.0.3       Ethernet0/0
4.4.4.4          1     FULL/DR         00:00:36    50.0.0.4       Ethernet0/0
R1#
```